

Amplified Carousel: Amplifying the Perception of Vertical Movement using Optical Illusion

Mondo Saito
mondo.saito@keio.jp
Keio University
Fujisawa, Kanagawa, Japan

Yasuto Nakanishi
Keio University
Fujisawa, Kanagawa, Japan
naka@sfc.keio.ac.jp

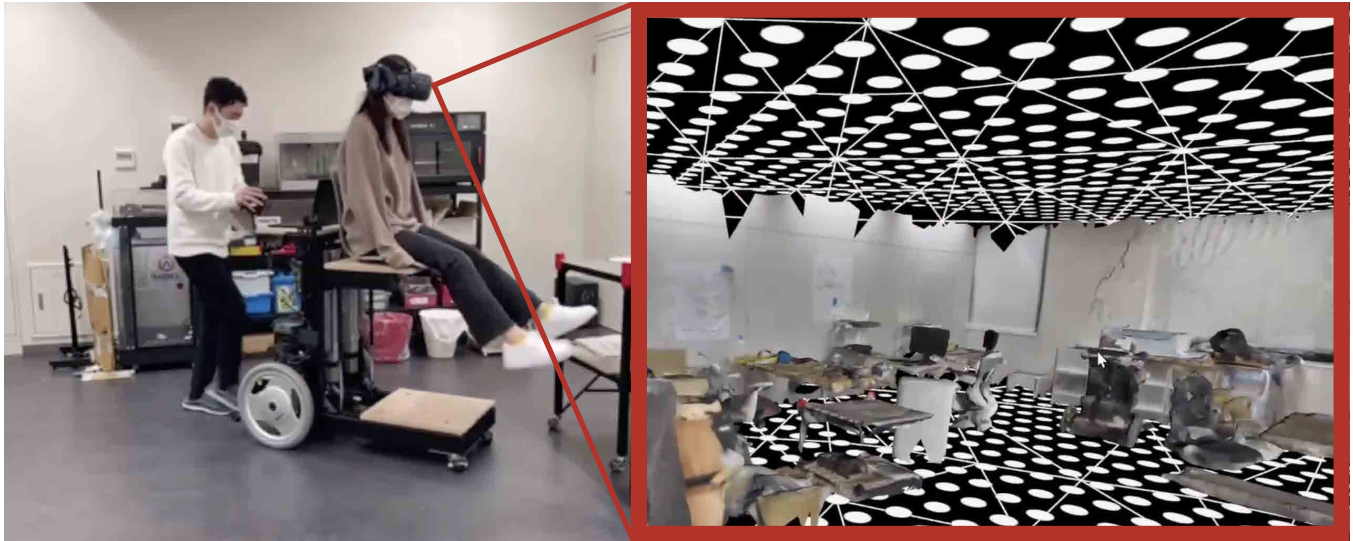


Figure 1: Vertical acceleration presenter (left) and view from the head mounted display of the optical illusion displayed on the floor and ceiling (right).

ABSTRACT

With the spread of virtual reality (VR) attractions, vector generation techniques that enhance the sense of realism are gaining attention. Additionally, mixed reality (MR) attractions, which overlay VR onto a real-world display, are expected to become more prevalent in the future. However, with MR, it is impossible to move all the coordinates of the visual stimuli to generate propervection effects. Therefore, we have created an optical illusion method that provides a three-dimensional impression of a two-dimensional visual stimulation. The technique amplifies the sensation of vertical movement by placing the illusion on the floor.

CCS CONCEPTS

• Human-centered computing → User interface design; User interface toolkits.

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KEYWORDS

vection, Optical illusion, Virtual Reality, Mixed Reality, Motion ride

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1 INTRODUCTION

With the recent increase in motion-ride facilities using virtual reality (VR) technology, research on visually guided self-motion sensation (i.e.,vection) systems has attracted attention in terms of improving the immersive experience. Vection provides motion sensations obtained by observing visual stimuli presented over a wide viewing area with physical movements in the same direction. For example, when sitting in a stationary vehicle, and a truck on either side begins moving forward, one may feel like the vehicle in which they are sitting is moving backward. Thus,vection is a phenomenon in which motion is perceived in the opposite direction of the visual stimulus. The technology used to amplify the intensity of vertical linearvection (LV) in MR motion rides is indispensable

for designing extraordinary virtual experiences with sensations of rising and dropping.

The main problem is that a large amount of energy is required to give a person vertical iso-accelerative motion in response to visual stimuli. Another problem is that the longer the duration of iso-accelerative motion, the greater the acceleration required to provide the participant a sensation of moving a long distance. Such devices express pseudo-vertical acceleration by repeatedly moving the viewer forward and backward [5]. Notably, the real acceleration takes place in horizontal linear directions. This subterfuge is made possible because the human visual acceleration perception characteristic dominates the vestibular system [2].

On the other hand, although it is possible to allow the participant to freely move while presenting all coordinates of visual stimuli in a VR space, it is not possible for a motion-riding mixed reality (MR) application because virtual objects are superimposed on the surrounding real-world scenery. Hence, utilizing the human acceleration perception characteristics described above is difficult.

Our proposed system provides a pseudo presentation of vertical LV using visual illusions in the MR space. Some types of optical illusions can express three-dimensional impressions in a two-dimensional projection, and by placing an illusion on a floor surface, the sensation of vertical movement can be stimulated and amplified. As a preliminary study, we investigated the effects of presenting such illusions in a VR space to emulate movements in the vertical direction. Three types of illusions were applied: amplification rate of vertical LV, horizontal angle of gaze, and their placement positions.

2 SYSTEM

2.1 VR system

Johansson et al. reported that the wider the presentation surface of visual stimuli, the stronger the LV occurrence based on the relationship between the perceived intensity of vection and the user's peripheral vision [3]. Therefore, HTC's VIVE PRO2, which has a relatively wide field of view among head-mounted displays (HMDs) was used as the HMD for the experiment. It presents visual stimuli at 120° horizontal and vertical ranges and has a refresh rate of 120 Hz.

2.2 Vertical acceleration presenter

To move the participants vertically, we created a mobile robot equipped with two Oriental Motor electric sliders (EASM6NYE030AZMC) that control acceleration and movement (Figure 2). We installed an emergency stop button to disconnect the main power in case of emergencies or any feelings of peril or discomfort expressed by the participant. The unit has a travel distance of 200 mm in the vertical direction. We controlled the acceleration and travel distance of the motorized slider using dedicated software (MEXE02). Figure 3 shows the results of a single acceleration cycle (i.e., descent and ascent) of the seat surface from rest. The number of motor revolutions was measured. Movement is "positive" when the seat surface rises and "negative" when it drops.

Hansen et al. reported that when presenting a visual stimulus of riding inside a VR moving vehicle, simulated wheel vibrations reduced VR sickness and increased the sense of immersion [1]. In

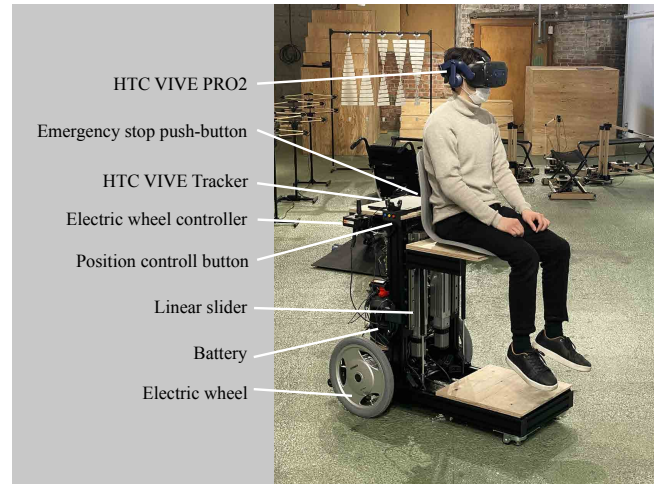


Figure 2: Vertical acceleration presenter

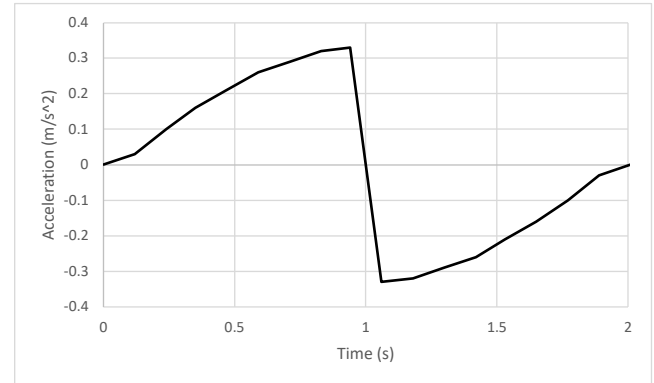


Figure 3: Acceleration of the vertical acceleration presenter as it raises and drops.

our system, vibrations generated during movement were provided by a YAMAHA electric wheelchair (TOWNYJOY X PLUS+).

Background music was also applied during the operation of the MR merry-go-round using the HMD speakers. We reduced the influence of the sound of the motor operation on the sensation of movement. Because recent research [6] has indicated that vection intensity can change depending on the meaning of the stimulus, we aimed for a meaning that would not cause discomfort from the combined visual and auditory information.

2.3 Visual stimulus

Three types of dynamic illusions were examined in their induction of vertical LV (Figure 4). To identify the one that produces the strongest vertical LV. We investigated the intensity of vection perceived with each illusion separately. We instructed participants to stand and look at their feet (Figure 5).

(a) Illusion of Y-junctions

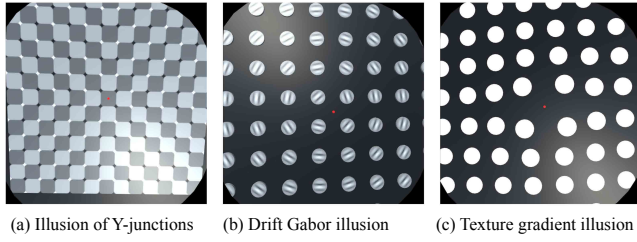


Figure 4: Three types of visual stimuli used in Experiments

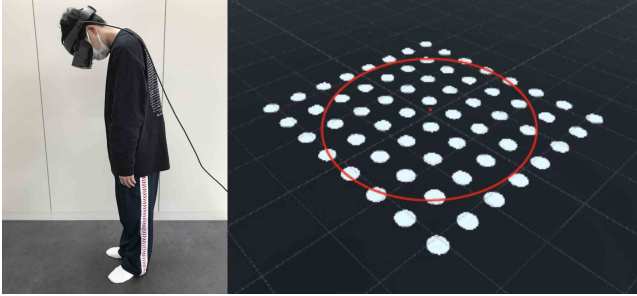


Figure 5: The posture that we instructed to participants in Experiments and visual stimulus in VR space.

We generated the Y-junction moving illusion created by Kitaoka [4], which creates a three-dimensional bulging illusion using alternating black-and-white star patterns at the vertices of a checkerboard square. We varied the intensity of the illusion by changing the grayscale settings of the star pattern. The checkerboard pattern was created by dividing the 8-bit grayscale into three parts and using the colors of the two intermediate points.

(b) Drift Gabor illusion

The Drift Gabor illusion is based on the Curveball illusion created by Shapiro et al. [8]. The circle moves perpendicular to the other stripes, creating the illusion that it is moving in a direction other than the actual direction. The color of the floor surface and the background of the Gabor patch is the color of the midpoint of the 8-bit grayscale divided into two parts.

(c) Texture gradient illusion

The texture gradient illusion simulates a depression forming on the floor. Concavity is expressed by increasing the perceived movement of the circle toward the user. The closer the circle, the more it seems to move along a radiation line centered on the user's feet. We can change the intensity of the three-dimensional effect by adjusting the perceived amount of movement. The color of the floor surface and the background of the circles are black. Figure 6 shows the shape of the Texture gradient illusion when raising and dropping.

3 EVALUATION AND FUTURE WORK

3.1 Evaluation

Four user studies were conducted on illusions that amplify the sense of vertical movement [7], which works by displaying a texture gradient illusion on the floor while the viewer gazes in a direction that

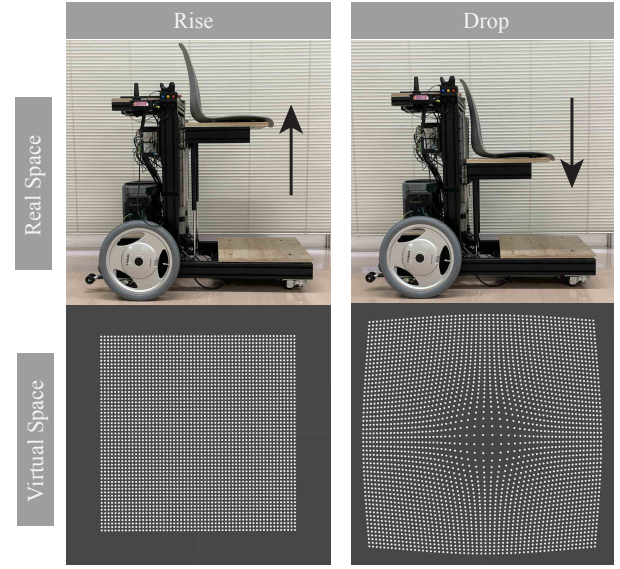


Figure 6: The shape of the Texture gradient illusion when raising and dropping.

places the illusion in the periphery. However, the scale of the user survey ($n=10$) limited the experiment's reliability. Therefore, we plan to increase the number of participants and conduct additional validation tests.

3.2 Future work

In the future, we plan to explore this system's applicability to amplifying the sense of horizontal movements. Based on the knowledge potentially gained from the combined amplification of vertical and horizontal movement sensations using illusory VR projections, we plan to extend the system to a mobile carousel MR application that enhances safety by creating the illusion of large movements with small movements.

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REFERENCES

- [1] John Paulin Hansen, Astrid Kofod Trudslev, Sara Amdt Harild, Alexandre Alapetite, and Katsumi Minakata. 2019. Providing Access to VR Through a Wheelchair. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (CHI EA '19). Association for Computing Machinery, New York, NY, USA, 1–8. <https://doi.org/10.1145/3290607.3299048>
- [2] L. Harris, M. Jenkin, and D.C. Zikovitz. 1999. Vestibular cues and virtual environments: choosing the magnitude of the vestibular cue. In *Proceedings IEEE Virtual Reality (Cat. No. 99CB36316)*. Institute of Electrical and Electronics Engineers, New York, NY, USA, 229–236. <https://doi.org/10.1109/VR.1999.756956>
- [3] Gunnar Johansson. 1977. Studies on Visual Perception of Locomotion. *Perception* 6 (1977), 365–376.
- [4] Akiyoshi Kitaoka, Baingio Pinna, and Gavin Brelstaff. 2004. Contrast Polarities Determine the Direction of Café Wall Tilts. *Perception* 33 (2004), 11–20.
- [5] Ryo Kodama, Masahiro Koge, Shun Taguchi, and Hiroyuki Kajimoto. 2017. COMS-VR: Mobile virtual reality entertainment system using electric car and head-mounted display. In *2017 IEEE Symposium on 3D User Interfaces (3DUI)*. Institute of Electrical and Electronics Engineers, New York, NY, USA, 130–133.

- <https://doi.org/10.1109/3DUI.2017.7893329>
- [6] Masaki Ogawa and Takeharu Seno. 2014. Vection is Modulated by the Semantic Meaning of Stimuli and Experimental Instructions. *Perception* 43 (09 2014), 605–15. <https://doi.org/10.1068/p7639>
- [7] Mondo Saito and Yasuto Nakanishi. 2022. Amplification of Vertical Acceleration using Optical Illusion. *IPSJ Technical Report* 63, 2 (02 2022), 437–445. <https://doi.org/10.20729/00216251>
- [8] Arthur Shapiro, Zhong-Lin Lu, Chang-Bing Huang, Emily Knight, and Robert Ennis. 2010. Transitions between Central and Peripheral Vision Create Spatial/Temporal Distortions: A Hypothesis Concerning the Perceived Break of the Curveball. *PLoS one* 5 (10 2010), e13296. <https://doi.org/10.1371/journal.pone.0013296>